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Tests for high-dimensional covariance matrices using the theory of U-statistics

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Test statistics for sphericity and identity of the covariance matrix are presented, when the data are multivariate normal and the dimension, p, can exceed the sample size, n. Under certain mild conditions mainly on the traces of the unknown covariance matrix, and using the asymptotic theory of U-statistics, the test statistics are shown to follow an approximate normal distribution for large p, also when $p \gg n$. The accuracy of the statistics is shown through simulation results, particularly emphasizing the case when p can be much larger than n. A real data set is used to illustrate the application of the proposed test statistics.

Keywords: covariance testing; high-dimensional data; sphericity; U-statistics

1. Introduction

The need for estimation and testing of high-dimensional covariance matrix in multivariate set-up has recently been galvanized by frequently encountered large data sets, particularly, but not necessarily limited to, genetics, microarray and astronomy. The present manuscript focuses on two tests of covariance matrix when the data are high dimensional. Precisely, suppose

$$\mathbf{X}_k = (X_{k1}, \ldots, X_{kp})' \sim \mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}),$$

$k = 1, \ldots, n$, are n independent and identically distributed random vectors, where $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ denote the mean vector and the covariance matrix, respectively. The objective is to test the hypotheses

$$H_{01} : \boldsymbol{\Sigma} = \sigma^2\mathbf{I} \quad \text{vs.} \quad H_{11} : \boldsymbol{\Sigma} \neq \sigma^2\mathbf{I}$$

and

$$H_{02} : \boldsymbol{\Sigma} = \mathbf{I} \quad \text{vs.} \quad H_{12} : \boldsymbol{\Sigma} \neq \mathbf{I},$$

particularly when $p > n$, where $\mathbf{I}$ is the identity matrix, and $\sigma^2 > 0$ is some constant. The first hypothesis, H_{01}, refers to the well-known sphericity hypothesis, whereas H_{02} is a convenient representation of a more general hypothesis $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}_0$, where $\boldsymbol{\Sigma}_0$ is any known positive-definite covariance matrix.

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There has been some interesting work on the tests of H_{01} and H_{02}, under high-dimensional set-up, over the last few years. Ledoit and Wolf [1] discuss the validity of the test statistics

$$U = \frac{1}{p}\,\mathrm{tr}\left(\frac{\mathbf{S}}{(1/p)\,\mathrm{tr}(\mathbf{S})} - \mathbf{I}\right)^2, \quad \text{and} \quad V = \frac{1}{p}\,\mathrm{tr}(\mathbf{S} - \mathbf{I})^2, \tag{1}$$

for H_{01} and H_{02}, respectively, under high-dimensional settings, where $\mathbf{S}$ is the sample estimator of $\boldsymbol{\Sigma}$, and tr denotes the trace. For $n \to \infty$ and p fixed, i.e. under n-asymptotics, and assuming normality, John [2] proposed the statistic U for H_{01} and proved it to be locally most powerful invariant, and Nagao [3] proposed the statistic V for H_{02}.

Ledoit and Wolf examine the behaviour of U and V when $p > n$, under some additional assumptions, particularly assuming that $p/n \to c \in (0, \infty)$. They show that, under normality, a test based on U is still consistent, even if $p > n$, but the same is not true for V. They then propose a modified version of V which is valid to test H_{02} under new assumptions; see also Fujikoshi et al. [4], Ch. 8. Birke and Dette [5] extend the work of Ledoit and Wolf by considering the same test statistics for the extreme boundaries of concentration, i.e. when $p/n \to c \in [0, \infty]$. Birke and Dette show that the statistics are also valid for extreme cases, although their approximating normal distributions needed to be derived through an approach different from the usual delta method employed by Ledoit and Wolf.

Under similar assumptions on the traces of the covariance matrix, Srivastava [6] proposed test statistics for spherical, identity, and diagonal covariance matrix. Assuming normality, the test statistics are shown to asymptotically follow a normal distribution when $p > n$. The robustness of the same test statistics is then evaluated in [7] under more strict assumptions on the traces, and putting restrictions on the moments of the underlying distribution. Chen et al. [8] have recently proposed tests for H_{01} and H_{02}, assuming vanishing trace ratios of the unknown covariance matrix. They do not assume normality, but place several restrictions, close to normality, on the moments of the underlying multivariate model.

In the sections to follow, new test statistics for H_{01} and H_{02} are proposed under a few mild conditions. The accuracy of the statistics is shown to remain intact under the assumed conditions even when p is much larger than n. The asymptotic normality of the proposed test statistics is derived using the theory of U-statistics. The rest of the article is organized as follows.

The test statistics and their limiting distributions are given in the next section. The performance of the statistics is shown through simulation studies in Section 3, where an application of sphericity test on a cancer data set is illustrated in Section 4. Section 5 summarizes the results, and main theoretical derivations are collected in the appendix.

2. The proposed test statistics

Let $\mathbf{X}_k = (X_{k1}, \ldots, X_{kp})' \sim \mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\boldsymbol{\Sigma} > 0$, $k = 1, \ldots, n$, be the model, as stated above. For the sequel of computations that follows, we assume $\boldsymbol{\mu} = \mathbf{0}$.

Consider the hypotheses H_{01} and H_{02}, and the corresponding statistics U and V in Equation (1). They are computed using the sample covariance matrix $\mathbf{S}$ as a plug-in estimator of $\boldsymbol{\Sigma}$ in

$$\frac{1}{p}\,\mathrm{tr}\left(\frac{\boldsymbol{\Sigma}}{(1/p)\,\mathrm{tr}(\boldsymbol{\Sigma})} - \mathbf{I}\right)^2 = \frac{(1/p)\,\mathrm{tr}(\boldsymbol{\Sigma}^2)}{[(1/p)\,\mathrm{tr}(\boldsymbol{\Sigma})]^2} - 1 = \frac{p\,\mathrm{tr}(\boldsymbol{\Sigma}^2)}{[\mathrm{tr}(\boldsymbol{\Sigma})]^2} - 1 \tag{2}$$

$$\text{and} \quad \frac{1}{p}\,\mathrm{tr}(\boldsymbol{\Sigma} - \mathbf{I})^2 = \frac{1}{p}\,\mathrm{tr}(\boldsymbol{\Sigma}^2) - \frac{2}{p}\,\mathrm{tr}(\boldsymbol{\Sigma}) + 1, \tag{3}$$

respectively. To study the behaviour of Equations (2) and (3), we need estimators of $\text{tr}(\boldsymbol{\Sigma})$, $[\text{tr}(\boldsymbol{\Sigma})]^2$, and $\text{tr}(\boldsymbol{\Sigma}^2)$, and for the use of the statistics for high-dimensional set-up, the estimators must be consistent for large p, even if $p > n$. For $\mathbf{X}_k$, $k = 1, \ldots, n$, define $A_k = \mathbf{X}_k'\mathbf{X}_k$ as a quadratic form, and $A_{kl} = \mathbf{X}_k'\mathbf{X}_l$, $k \neq l$, as a symmetric bilinear form. The estimators of the three traces, $\text{tr}(\boldsymbol{\Sigma})$, $[\text{tr}(\boldsymbol{\Sigma})]^2$, and $\text{tr}(\boldsymbol{\Sigma}^2)$, are given in the following definition.

Definition 1 *Let A_k and A_{kl} be as defined above. Then*

$$C_1 = \frac{1}{n}\sum_{k=1}^{n} A_k, \tag{4}$$

$$C_2 = \frac{1}{n(n-1)} \underbrace{\sum_{k=1}^{n}\sum_{l=1}^{n}}_{k\neq l} A_k A_l, \tag{5}$$

$$C_3 = \frac{1}{n(n-1)} \underbrace{\sum_{k=1}^{n}\sum_{l=1}^{n}}_{k\neq l} A_{kl}^2. \tag{6}$$

are the estimators of $\text{tr}(\boldsymbol{\Sigma})$, $[\text{tr}(\boldsymbol{\Sigma})]^2$, *and* $\text{tr}(\boldsymbol{\Sigma}^2)$, *respectively.*

Note that, C_1, C_2 and C_3 are moment estimators of the respective traces, as opposed to the plug-in estimators using sample covariance matrix $\mathbf{S}$, given in Equation (1). The desired properties of the estimators are established in Lemma 2. The estimators are already discussed in [9, Ch. 2] in the context of presenting a statistic for mean testing in high-dimensional longitudinal data, where it is proved that the estimators are unbiased and consistent, and the consistency remains intact even if the dimension exceeds the sample size; see also Ahmad et al.[10]

The following two results, Lemma 2 and Corollary 3, proved in Appendix 2, summarize the basic properties of the estimators given in Definition 1.

Lemma 2 *Let the estimators C_1, C_2 and C_3, be as given in Definition 1. Then*

$$\text{E}(C_1) = \text{tr}(\boldsymbol{\Sigma}), \quad \text{Var}(C_1) = \frac{2}{n}\,\text{tr}(\boldsymbol{\Sigma}^2),$$

$$\text{E}(C_2) = [\text{tr}(\boldsymbol{\Sigma})]^2, \quad \text{Var}(C_2) = \frac{8}{n(n-1)}[(n-1)\,\text{tr}(\boldsymbol{\Sigma}^2)[\text{tr}(\boldsymbol{\Sigma})]^2 + \left[\text{tr}(\boldsymbol{\Sigma}^2)\right]^2]$$

$$\text{E}(C_3) = \text{tr}(\boldsymbol{\Sigma})^2, \quad \text{Var}(C_3) = \frac{4}{n(n-1)}[(2n-1)\,\text{tr}(\boldsymbol{\Sigma}^4) + [\text{tr}(\boldsymbol{\Sigma}^2)]^2],$$

$$\text{Cov}(C_1, C_2) = \frac{4}{n}\,\text{tr}(\boldsymbol{\Sigma})\,\text{tr}(\boldsymbol{\Sigma}^2),$$

$$\text{Cov}(C_1, C_3) = \frac{4}{n}\,\text{tr}(\boldsymbol{\Sigma}^3),$$

$$\text{Cov}(C_2, C_3) = \frac{8}{n(n-1)}[\text{tr}(\boldsymbol{\Sigma}^4) + (n-1)\,\text{tr}(\boldsymbol{\Sigma})\,\text{tr}(\boldsymbol{\Sigma}^3)].$$

COROLLARY 3 *Given Lemma 2. Then, for $i \neq j$,*

$$\mathrm{Var}\left(\frac{C_i}{\mathrm{E}(C_i)}\right) = O\left(\frac{1}{n}\right),$$

$$\mathrm{Cov}\left(\frac{C_i}{\mathrm{E}(C_i)}, \frac{C_j}{\mathrm{E}(C_j)}\right) = O\left(\frac{1}{n}\right),$$

where $i, j = 1, 2, 3$, i.e. the variance- and covariance-ratios are uniformly bounded in p.

Corollary 3 is an immediate consequence of Lemma 2 and, hence, its proof, in Appendix 2, is embedded in the proof of Lemma 2. But it is separately stated because of its value in proving the high-dimensional consistency of the test statistics. Of particular use is the boundedness of the variance ratio of C_3 since it will help us establish non-degenerate limit distribution of the test statistics using the square integrability of the kernels of the U-statistics.

In the next two sections, the test statistics for H_{01} and H_{02} are constructed, based on Definition 1 and the results of Lemma 2. We begin with the sphericity test.

2.1. *Test statistic for H_{01}*

We use estimators given in Definition 1 to define the estimator of Equation (2) as

$$T_1 = \frac{pC_3}{C_2} - 1 = \hat{\psi} - 1, \tag{7}$$

where $\hat{\psi} = pC_3/C_2$ estimates $\psi = p\,\mathrm{tr}(\mathbf{\Sigma}^2)/[\mathrm{tr}(\mathbf{\Sigma})]^2$. To compute the asymptotic distribution of T_1, we need to compute the moments of T_1. As T_1 involves a ratio of two correlated estimators, exact moments of T_1 cannot be computed, where an approximation of the first two moments can be obtained using the bivariate delta method of moments [11, p Ch. 5]; see also Ahmad [9] and other related references cited therein.

Since, our main focus is on the application of T_1 under high-dimensional set-up, we approach the problem of asymptotic normality from a slightly different, and relatively simpler way. First, we note that C_1, C_2 and C_3 are natural representations of U-statistics.[12] Furthermore, by Lemma 2, the variance-ratios of the estimators are uniformly bounded in p. This helps us show the asymptotic result differently. For this, we re-write T_1 as

$$\frac{T_1 + 1}{\psi} = \frac{pC_3}{C_2} \cdot \frac{[\mathrm{tr}(\mathbf{\Sigma})]^2}{p\,\mathrm{tr}(\mathbf{\Sigma}^2)} = \frac{C_3/\mathrm{tr}(\mathbf{\Sigma}^2)}{C_2/[\mathrm{tr}(\mathbf{\Sigma})]^2}. \tag{8}$$

Using the results of Lemma 2, it can be easily shown that

$$\frac{C_2}{[\mathrm{tr}(\mathbf{\Sigma})]^2} \xrightarrow{\mathcal{P}} 1,$$

where $\xrightarrow{\mathcal{P}}$ denotes convergence in probability. Then, we only need to show the asymptotic normality of the numerator of Equation (8) which, from Definition 1, can be written as

$$\frac{T_1 + 1}{\psi} = \frac{C_3}{\mathrm{tr}(\mathbf{\Sigma}^2)}[1 + o_P(1)]^{-1} = U_n[1 + o_P(1)]^{-1}, \tag{9}$$

with

$$U_n = \frac{1}{n(n-1)} \sum_{k=1}^{n} \sum_{\substack{l=1 \\ k \neq l}}^{n} \frac{A_{kl}^2}{\mathrm{tr}(\mathbf{\Sigma}^2)}, \tag{10}$$

where U_n is in the form of a U-statistic.[13,14] We, therefore, use the asymptotic theory of U-statistics to prove the asymptotic normality of T_1. The theory of U-statistics, in particular asymptotic limit distribution, has been extensively investigated under a variety of frameworks, as can be witnessed from book-length references cited above, in particular the later reference with its rich bibliography. Further short but comprehensive treatment can be found in [15, Ch. 6], [16, Ch. 5] and [17, Ch. 12]. In particular, we shall be using Theorem 6.1.2(ii) in [15, p.369], heretofore referred to as Lehmann's theorem, to fix the asymptotic distribution of U_n of Equation (10).

It may well be re-emphasized here that Lehmann's theorem is applicable for the proposed test statistics even when $p \to \infty$. The non-degenerate asymptotic normal limit in Lehmann's theorem rests on the behaviour of the component variances (of projected kernels), ξ_c, in $\mathrm{Var}(U_n)$. In particular, it needs to be ensured that $\xi_c\ \forall c$ (in the present case, $c = 1, 2$), is uniformly bounded away from both 0 and ∞. For the treatment of degenerate kernels in similar contexts under different conditions of degeneracy, see Ahmad et al., [18] and Ahmad.[19] Now, for valid application of Lehmann's theorem, we need to set the following assumptions.

Assumption 4 For $p \to \infty$, let $\mathrm{tr}(\boldsymbol{\Sigma})/p = O(1)$.

Assumption 5 $\mathrm{tr}(\boldsymbol{\Sigma}^2)/p^2 = \delta$, where $0 < \delta \leq \infty$.

Assumption 6 For $n, p \to \infty$, $p/n \to c \in (0, \infty)$.

Assumption 7 $\inf_p(\mathrm{tr}(\boldsymbol{\Sigma}^4)/[\mathrm{tr}(\boldsymbol{\Sigma}^2)]^2) > 0$.

Note that, Assumption 4 is quite logical and needs no serious rationale, and Assumption 5 is specifically aimed to cover most of the practically used covariance structures, e.g. compound symmetry. It must also be noted here, that Assumption 5 is specifically taken for the distribution of the statistic under the alternative. It is not needed under the null, when it is obviously not satisfied, and the last two assumptions are needed to ensure non-degenerate limits of T_1 and T_2 under the null and alternative hypotheses, respectively. Now, to use Lehmann's theorem for T_1, we need the moments of T_1. Under the aforementioned assumptions, the moments of the T_1 can be approximated (ignoring the terms that vanish in limit), using the results in Lemma 2, as

$$\mathrm{E}\left(\frac{T_1 + 1}{\psi}\right) = 1, \tag{11}$$

$$\mathrm{Var}\left(\frac{T_1 + 1}{\psi}\right) = \frac{4}{n(n-1)}\left[\frac{(2n-1)\,\mathrm{tr}(\boldsymbol{\Sigma}^4)}{\left[\mathrm{tr}(\boldsymbol{\Sigma}^2)\right]^2} + 1\right], \tag{12}$$

which, under H_0, reduce to the simpler forms

$$\mathrm{E}(T_1) = 0, \tag{13}$$

$$\mathrm{Var}(T_1) = \frac{4}{n(n-1)}\left(\frac{2n-1}{p} + 1\right), \tag{14}$$

so that, nT_1 has a finite limit as $n, p \to \infty$, under H_0 and under the assumptions stated above. With this set-up, an application of Lehmann's theorem for $m = 2$ gives the asymptotic normal distribution of U_n in Equation (10), and hence, of T_1, using Slutsky's theorem [20, Theorem 2.13, p.30]. We state the following theorem, proved in Appendix 3.

Theorem 8 *Let T_1 be as defined in Equation (7). Then, under Assumptions 4–7,*

$$\sigma_{T_1}^{-1}\left(\frac{T_1+1}{\psi}-1\right)\xrightarrow{\mathcal{D}}\mathrm{N}(0,1), \tag{15}$$

as $p,n\to\infty$, where $\sigma_{T_1}^2$ is the asymptotic variance of T_1 given in Equation (12).

In particular, under the null hypothesis, we have the following simple form of the test statistic.

Corollary 9 *Given Theorem 8. Then, under H_{01} and Assumptions 4–6,*

$$nT_1\xrightarrow{\mathcal{D}}\mathrm{N}\left(0,4\left(\frac{2}{c+1}\right)\right),$$

as $p,n\to\infty$.

2.2. *Test statistic for H_{02}*

Following the same lines, to compute the test statistic for H_{02}, we plug in the relevant estimators from Definition 1 into Equation (3), and get

$$T_2=\frac{1}{p}C_3-\frac{2}{p}C_1+1. \tag{16}$$

Clearly, T_2 is an unbiased estimator of Equation (3), or pT_2 is an unbiased estimator of $\operatorname{tr}(\mathbf{\Sigma}-\mathbf{I})^2$. As T_2 is simply a linear combination of the estimators, computation of the moments of T_2 is trivial, and the bivariate delta method of moments can be used to fix the asymptotic distribution of T_2, as argued for T_1 above. But, a closer look at Equation (16) clues to the fact that the asymptotic normality of T_2 can be shown using merely the results of Section 2.1, without any new computations. We write,

$$\begin{aligned}T_2-\frac{1}{p}\operatorname{tr}(\mathbf{\Sigma}-\mathbf{I})^2&=\frac{1}{p}[C_3-\operatorname{tr}(\mathbf{\Sigma}^2)]-\frac{2}{p}[C_1-\operatorname{tr}(\mathbf{\Sigma})]\\&=\frac{\operatorname{tr}(\mathbf{\Sigma}^2)}{p}\left(\frac{C_3}{\operatorname{tr}(\mathbf{\Sigma}^2)}-1\right)-\frac{2\operatorname{tr}(\mathbf{\Sigma})}{p}\left(\frac{C_1}{\operatorname{tr}(\mathbf{\Sigma})}-1\right)\end{aligned} \tag{17}$$

We follow the same strategy as for T_1. From Lemma 2, it can be shown that

$$\frac{C_1}{\operatorname{tr}(\mathbf{\Sigma})}\xrightarrow{\mathcal{P}}1,$$

so that we can write

$$\begin{aligned}T_2-\frac{1}{p}\operatorname{tr}(\mathbf{\Sigma}-\mathbf{I})^2&=\frac{\operatorname{tr}(\mathbf{\Sigma}^2)}{p}\left(\frac{C_3}{\operatorname{tr}(\mathbf{\Sigma}^2)}-1\right)-2O(1)[o_P(1)]\\&=\frac{\operatorname{tr}(\mathbf{\Sigma}^2)}{p}(U_n-1)+o_P(1),\end{aligned} \tag{18}$$

where U_n is as the same as defined in Equation (10) for T_1. Therefore, the asymptotic normality of U_n, as proved for T_1, combined again with Slutsky's theorem, can be used to prove asymptotic normality of T_2. Furthermore, the approximate moments, ignoring the constants that vanish in limit, can be expressed as

$$\mathrm{E}\left(T_2 - \frac{1}{p}\,\mathrm{tr}(\mathbf{\Sigma} - \mathbf{I})^2\right) = 0, \tag{19}$$

$$\mathrm{Var}\left(T_2 - \frac{1}{p}\,\mathrm{tr}(\mathbf{\Sigma} - \mathbf{I})^2\right) = \frac{4[\mathrm{tr}(\mathbf{\Sigma}^2)]^2}{n(n-1)p^2}\left[\frac{(2n-1)\,\mathrm{tr}(\mathbf{\Sigma}^4)}{[\mathrm{tr}(\mathbf{\Sigma}^2)]^2} + 1\right], \tag{20}$$

which, under H_{02}, reduce to those under H_{01} as given in Equations (13) and (12), respectively, so that, likewise, nT_2 also has a finite limit under H_{02}. We summarize the results in the following theorem on the asymptotic distribution of T_2, proved in Appendix 3.

Theorem 10 *Let T_2 be as defined in Equation (7). Then, under Assumptions 4–7,*

$$\sigma_{T_2}^{-1}\left[T_2 - \frac{1}{p}\,\mathrm{tr}(\mathbf{\Sigma} - \mathbf{I})^2\right] \xrightarrow{\mathcal{D}} \mathrm{N}(0, 1), \tag{21}$$

as $p, n \to \infty$, where $\sigma_{T_2}^2$ is the asymptotic variance of T_2 given in Equation (12).

Further, under the null hypothesis, we have the following simple form of the test statistic.

Corollary 11 *Given Theorem 10. Then, under H_{02} and Assumptions 4–6,*

$$nT_2 \xrightarrow{\mathcal{D}} \mathrm{N}\left(0, 4\left(\frac{2}{c+1}\right)\right),$$

as $p, n \to \infty$.

We note that, the distributions of the two test statistics are same because of the same scaled kernel used to approximate their composition in Equations (9) and (18). In the next section, it is verified through simulations, that the approximating distributions are essentially developed for $p \to \infty$, disregarding how large n is. Although, the asymptotic theory of U-statistics requires the condition $n \to \infty$, but as clearly shown in the simulated results in Section 3, the accuracy of the statistics is not disturbed even for moderate n, and for a very large p.

3. Simulation results

Table 1 reports estimated quantiles for T_1 (upper panel) and T_2 (lower panel) for different pairs of n and p, and for three nominal quantiles 0.9, 0.95 and 0.99. The small-n, large-p accuracy of both test statistics can be clearly evidenced from the estimated values for $n = 5$ where $p = 1000$. As expected, the accuracy of the statistics increases for increasing n, against any p, but this accuracy is not damaged for any fixed n when p is allowed to increase. This validates the high-dimensional consistency of the estimators used to construct the test statistics. We observe an increasing stability of the estimated quantiles for increasing n and p, where T_1 slightly outperforms T_2.

Table 2 reports power values for the two tests for the same pairs of n and p as used for quantile estimation, and under three most commonly assumed covariance structures for multivariate analysis, viz. compound symmetry, first-order autoregressive, and unstructured (UN), where the nominal quantile is fixed at 0.95. A compound symmetric (CS) covariance structure is defined as $\mathbf{\Sigma} = \sigma^2\mathbf{I} + \rho\mathbf{J}$ where $\mathbf{I}$ is the identity matrix and $\mathbf{J}$ is the matrix of ones. A covariance structure

Table 1. Estimated test size for T_1 (Equation (7)) and T_2 (Equation (16)).

(T_1)		p							
n	$1-\alpha$	5	10	20	50	100	200	500	1000
5	0.90	0.917	0.909	0.900	0.892	0.893	0.920	0.890	0.891
	0.95	0.950	0.945	0.940	0.962	0.956	0.954	0.957	0.958
	0.99	0.983	0.980	0.987	0.984	0.983	0.983	0.982	0.984
10	0.90	0.913	0.900	0.900	0.906	0.908	0.900	0.904	0.906
	0.95	0.949	0.959	0.956	0.951	0.956	0.950	0.951	0.953
	0.99	0.986	0.987	0.985	0.987	0.989	0.985	0.989	0.989
20	0.90	0.902	0.903	0.897	0.905	0.903	0.901	0.899	0.901
	0.95	0.958	0.952	0.951	0.953	0.952	0.950	0.954	0.949
	0.99	0.986	0.989	0.988	0.989	0.991	0.990	0.990	0.990
50	0.90	0.916	0.913	0.908	0.900	0.903	0.900	0.900	0.900
	0.95	0.954	0.955	0.954	0.952	0.95	0.950	0.952	0.949
	0.99	0.983	0.989	0.990	0.990	0.989	0.990	0.989	0.989
(T_2)		p							
n	$1-\alpha$	5	10	20	50	100	200	500	1000
5	0.90	0.921	0.911	0.915	0.902	0.895	0.895	0.895	0.894
	0.95	0.955	0.947	0.951	0.953	0.955	0.958	0.954	0.956
	0.99	0.977	0.977	0.980	0.982	0.982	0.983	0.981	0.982
10	0.90	0.916	0.909	0.900	0.905	0.908	0.904	0.904	0.905
	0.95	0.952	0.948	0.949	0.951	0.951	0.950	0.951	0.952
	0.99	0.980	0.978	0.983	0.986	0.986	0.988	0.987	0.989
20	0.90	0.910	0.908	0.904	0.901	0.906	0.902	0.902	0.903
	0.95	0.957	0.951	0.952	0.948	0.949	0.949	0.952	0.954
	0.99	0.984	0.984	0.987	0.988	0.990	0.988	0.991	0.991
50	0.90	0.919	0.902	0.902	0.900	0.900	0.898	0.899	0.900
	0.95	0.956	0.953	0.950	0.947	0.950	0.948	0.949	0.948
	0.99	0.984	0.985	0.989	0.988	0.990	0.988	0.992	0.989

is autoregressive of order 1, AR(1), if $\text{Cov}(X_k, X_l) = \sigma^2\rho^{|k-l|}$, $\forall k, l$, whereas the UN covariance matrix refers to $\mathbf{\Sigma} = (\sigma_{ij})_{i,j=1}^{p}$. For the computations reported in Table 2, it is assumed that $\sigma^2 = 1$, $\rho = 0.5$ for CS, $\sigma^2 = 1$, $\rho = 0.6$ for AR(1), and $\sigma_{ij} = 1(1)p$ $(i = j)$, $\rho_{ij} = (i-1)/p$ $(i > j)$, for UN.

It is observed that the power of both test statistics increases for increasing p for any n, and also for increasing n against any p. The highest power is generated by the CS covariance pattern, followed by autoregressive and UN patterns. In general, it can be concluded that the test statistics have high power for moderate n, say 10 or more, for any $p \lesseqgtr n$. It must be noted that the power pattern of the statistics for autoregressive covariance structure does not vary significantly by changing ρ. The assumed $\rho = 0.6$ was taken only as a moderate example value, but it was verified that similar results are produced by smaller or larger values of the correlation coefficient.

The autoregressive structure gives relatively low power, particularly for small sample size, and power increases slowly for increasing n and p, but still for n as moderate as 10 the power is high and increases for increasing dimension. The worst case scenario is the UN pattern for T_2 when $n = 5$ where power even goes down slightly. The performance, however, clearly rapidly improves for increasing n. The CS structure is the closest violation of the null hypothesis since any CS matrix can be orthonormally transformed to a spherical matrix. It can be generally concluded that the two statistics accurately control test size and have high power, and the accuracy increases with increasing dimension, for any n, particularly for $n \geq 10$.

Table 2. Power of test for T_1 (Equation (7)) and T_2 (Equation (16)) under the alternative covariance matrices CS, AR(1) and UN.

(T_1)		p							
$\mathbf{\Sigma}$	n	5	10	20	50	100	200	500	1000
CS	5	0.414	0.667	0.846	0.945	0.977	0.991	0.997	0.999
	10	0.759	0.937	0.989	1.000	1.000	1.000	1.000	1.000
	20	0.974	0.999	1.000	1.000	1.000	1.000	1.000	1.000
	50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
AR(1)	5	0.335	0.442	0.503	0.570	0.583	0.596	0.604	0.613
	10	0.707	0.844	0.907	0.955	0.967	0.970	0.970	0.972
	20	0.976	0.998	1.000	1.000	1.000	1.000	1.000	1.000
	50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
UN	5	0.131	0.167	0.191	0.205	0.224	0.222	0.220	0.223
	10	0.288	0.345	0.386	0.415	0.440	0.448	0.450	0.449
	20	0.607	0.734	0.814	0.853	0.870	0.882	0.888	0.892
	50	0.992	1.000	1.000	1.000	1.000	1.000	1.000	1.000
(T_2)		p							
$\mathbf{\Sigma}$	n	5	10	20	50	100	200	500	1000
CS	5	0.315	0.545	0.744	0.905	0.953	0.983	0.993	0.997
	10	0.650	0.894	0.980	0.999	1.000	1.000	1.000	1.000
	20	0.964	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
AR(1)	5	0.274	0.377	0.451	0.549	0.578	0.584	0.610	0.611
	10	0.578	0.762	0.867	0.940	0.956	0.967	0.971	0.973
	20	0.957	0.995	0.999	1.000	1.000	1.000	1.000	1.000
	50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
UN	5	0.024	0.020	0.017	0.016	0.016	0.014	0.014	0.013
	10	0.148	0.225	0.330	0.352	0.378	0.385	0.394	0.404
	20	0.843	0.982	0.998	1.000	1.000	1.000	1.000	1.000
	50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

4. Application

For real-life application of the test statistic, we apply T_1 to test sphericity for a leukaemia data set. The data are originally reported in [21] and is one of the most frequently used data set for high-dimensional analysis. The complete data consist of gene expressions for $p = 6817$ genes observed for two groups of sizes 47 (acute lymphoblastic leukaemia – ALL) and 25 (acute myeloid leukaemia – AML); see, e.g. Izenman.[22] For details on standardization and pre-processing of the data, see also Dudoit et al.[23] One part of the data set consist of 3051 genes on 38 tumour mRNA samples, with 27 in ALL and 11 in AML group, respectively. This part is also available in the R software repository under the package of `multtest`.[23]

For illustrative purposes, we only use a subset of this data comprising AML group with $n = 11$ and $p = 3051$, mainly to show the small sample performance of the test statistic. The close normal approximation of the data is first checked using QQ-plots. To test sphericity hypothesis, we compute $T_1 = 5.17$ with p-value 0, indicating a serious departure from the null hypothesis.

5. Summary and conclusions

Test statistics for sphericity and identity of a high-dimensional covariance matrix are presented under normality assumption. The statistics, based on unbiased and consistent estimators, follow approximate normal distribution and are also valid when the data are not high dimensional. The statistics are computed under mild conditions on the covariance matrix. The asymptotic normality of the test statistics is demonstrated using the theory of U-statistics.

Simulation results show that the statistics accurately control test size and have high power even when the dimension is much larger than the sample size. The power properties of the statistics are demonstrated to remain intact under a variety of alternative hypothesis. The general behaviour of the statistics is that they are accurate, both for size control and power, for a moderate sample size and large dimension. For relatively less complicated covariance structures, however, they are accurate even for a small sample size.

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Appendix 1. Some basic results

For the proof of Lemma 2, we need the following basic results.

THEOREM A.1 (Mathai and Provost [24, Ch. 3]; Mathai et al. [25, Ch. 2]) *Let* $\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, *and* $\mathbf{v} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$ $\boldsymbol{\Sigma} > 0$, *be* $p \times 1$ *random vectors, and* $\mathbf{A}$ *be any* $p \times p$ *symmetric matrix. Define* $Q = \mathbf{u}'\mathbf{A}\mathbf{u}$ *as a quadratic form, and* $B = \mathbf{u}'\mathbf{A}\mathbf{v}$ *as a symmetric bilinear form. Then, the rth cumulants of Q and B are given as*

$$\kappa_r(Q) = 2^{r-1}(r-1)!\,\mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma})^r, \quad r = 1, 2, \ldots,$$

$$\kappa_r(B) = \frac{1}{2}(r-1)!\,\mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma})^r, \quad r = 1, 2, \ldots,$$

respectively, where tr *denotes the trace. Particularly, for* $r = 1$ *and* 2, *we have*

$$E(Q) = \mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma}), \quad \mathrm{Var}(Q) = 2\,\mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma})^2$$

$$E(B) = 0, \quad \mathrm{Var}(B) = \mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma})^2,$$

assuming $\mathbf{u}$ *and* $\mathbf{v}$ *are independent (see also Ahmad* [9], *Appendix* 1).

LEMMA A.2 [26, Lemma 6.2, p. 209] *Let* $\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, $\boldsymbol{\Sigma} > 0$ *be a* $p \times 1$ *random vector, and* $\mathbf{A}$ *and* $\mathbf{B}$ *be* $p \times p$ *symmetric matrices. Define* $\mathbf{u}'\mathbf{A}\mathbf{u}$ *and* $\mathbf{u}'\mathbf{B}\mathbf{u}$ *be the quadratic forms. Then, the mean and variance of the product* $z = \mathbf{u}'\mathbf{A}\mathbf{u} \cdot \mathbf{u}'\mathbf{B}\mathbf{u}$ *are given as*

$$\begin{aligned}
\mathrm{E}(z) &= \mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma})\,\mathrm{tr}(\mathbf{B}\boldsymbol{\Sigma}) + 2\,\mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma}\mathbf{B}\boldsymbol{\Sigma}) \\
\mathrm{Var}(z) &= 32\,\mathrm{tr}[(\mathbf{A}\boldsymbol{\Sigma})^2(\mathbf{B}\boldsymbol{\Sigma})^2] \\
&\quad + 16[\mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma}\mathbf{B}\boldsymbol{\Sigma})^2 + \mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma})\,\mathrm{tr}\{(\mathbf{A}\boldsymbol{\Sigma})(\mathbf{B}\boldsymbol{\Sigma})^2\} + \mathrm{tr}(\mathbf{B}\boldsymbol{\Sigma})\,\mathrm{tr}\{(\mathbf{A}\boldsymbol{\Sigma})^2(\mathbf{B}\boldsymbol{\Sigma})\}] \\
&\quad + 4[\mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma})^2\,\mathrm{tr}(\mathbf{B}\boldsymbol{\Sigma})^2 + [\mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma}\mathbf{B}\boldsymbol{\Sigma})]^2 + \mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma})\,\mathrm{tr}(\mathbf{B}\boldsymbol{\Sigma})\,\mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma}\mathbf{B}\boldsymbol{\Sigma})] \\
&\quad + 2[[\mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma})]^2\,\mathrm{tr}(\mathbf{B}\boldsymbol{\Sigma})^2 + [\mathrm{tr}(\mathbf{B}\boldsymbol{\Sigma})]^2\,\mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma})^2].
\end{aligned}$$

Appendix 2. Proof of Lemma 2

The unbiasedness and consistency of the estimators is trivial (see also Ahmad [9], Ch. 2). Then, we work on the covariances. We have

$$\mathrm{Cov}(C_2, C_3) = \frac{1}{n^2(n-1)^2} \underbrace{\sum_{k=1}^{n}\sum_{l=1}^{n}}_{k \neq l} \underbrace{\sum_{r=1}^{n}\sum_{s=1}^{n}}_{r \neq s} \mathrm{Cov}(A_k A_l, A_{rs}^2), \tag{A1}$$

where, from Theorem A.1, $\mathrm{E}(A_k A_l) = [\mathrm{tr}(\boldsymbol{\Sigma})]^2$, $\mathrm{E}(A_{rs}^2) = \mathrm{tr}(\boldsymbol{\Sigma}^2)$, for $k \neq l$ and $r \neq s$. The covariance vanishes when $k \neq r, l \neq s$. The remaining cases are: $k = r, l = s; k = l, r \neq s; k \neq r, l = s; k \neq r, l \neq s$. For $k = l, r \neq s$

$$\begin{aligned}
\mathrm{E}(A_k A_l A_{ks}^2) &= \mathrm{E}(A_l)\mathrm{E}(A_k A_{ks}^2) = \mathrm{tr}(\boldsymbol{\Sigma})\mathrm{E}[\mathbf{X}_k'\mathbf{X}_k\mathbf{X}_k'\mathbf{X}_s\mathbf{X}_k'\mathbf{X}_s] \\
&= \mathrm{tr}(\boldsymbol{\Sigma})\mathrm{E}[\mathrm{tr}\{\mathbf{X}_k'\mathbf{X}_k\mathbf{X}_k'\mathbf{X}_s\mathbf{X}_s'\mathbf{X}_k\}] \\
&= \mathrm{tr}(\boldsymbol{\Sigma})\mathrm{E}[\mathrm{tr}\{\mathbf{X}_k\mathbf{X}_k'\mathbf{X}_k\mathbf{X}_k'\mathbf{X}_s\mathbf{X}_s'\}] \\
&= \mathrm{tr}(\boldsymbol{\Sigma})\,\mathrm{tr}[\mathrm{E}\{\mathbf{X}_k\mathbf{X}_k'\mathbf{X}_k\mathbf{X}_k'\}\boldsymbol{\Sigma}] = \mathrm{tr}(\boldsymbol{\Sigma})\mathrm{E}[\mathbf{X}_k'\mathbf{X}_k\mathbf{X}_k'\boldsymbol{\Sigma}\mathbf{X}_k] \\
&= \mathrm{tr}(\boldsymbol{\Sigma})[2\,\mathrm{tr}(\boldsymbol{\Sigma}^3) + \mathrm{tr}(\boldsymbol{\Sigma})\,\mathrm{tr}(\boldsymbol{\Sigma}^2)] \\
&= 2\,\mathrm{tr}(\boldsymbol{\Sigma})\,\mathrm{tr}(\boldsymbol{\Sigma}^3) + [\mathrm{tr}(\boldsymbol{\Sigma})]^2\,\mathrm{tr}(\boldsymbol{\Sigma}^2),
\end{aligned}$$

so that $\mathrm{Cov}(A_k A_l, A_{ks}^2) = 2\,\mathrm{tr}(\boldsymbol{\Sigma})\,\mathrm{tr}(\boldsymbol{\Sigma}^3)$. For $k = r, l = s$, write

$$A_{kl}^2(A_k + A_l)^2 = A_{kl}^2 A_k^2 + A_{kl}^2 A_l^2 + 2A_{kl}^2 A_k A_l,$$

so that

$$\mathrm{E}(A_k A_l A_{kl}^2) = \tfrac{1}{2}[\mathrm{E}[A_{kl}^2(A_k + A_l)^2] - \mathrm{E}(A_{kl}^2 A_k^2) - \mathrm{E}(A_{kl}^2 A_l^2)]. \tag{A2}$$

Let $\mathbf{Z} = (\mathbf{X}_k'\mathbf{X}_l')'$, with $\mathrm{Cov}(\mathbf{Z}) = \left(\begin{smallmatrix}\boldsymbol{\Sigma} & \mathbf{0}\\ \mathbf{0} & \boldsymbol{\Sigma}\end{smallmatrix}\right) = \mathbf{I}_2 \otimes \boldsymbol{\Sigma} = \mathbf{V}$(say). Then $A_{kl} = \frac{1}{2}\mathbf{Z}'\mathbf{A}\mathbf{Z}$, $A_k + A_l = \mathbf{Z}'\mathbf{B}\mathbf{Z}$, $A_k = \mathbf{Z}'\mathbf{C}\mathbf{Z}$, and $A_l = \mathbf{Z}'\mathbf{D}\mathbf{Z}$, where

$$\mathbf{A} = \begin{pmatrix}\mathbf{0} & \mathbf{I}\\ \mathbf{I} & \mathbf{0}\end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix}\mathbf{I} & \mathbf{0}\\ \mathbf{0} & \mathbf{I}\end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix}\mathbf{I} & \mathbf{0}\\ \mathbf{0} & \mathbf{0}\end{pmatrix}, \quad \text{and} \quad \mathbf{D} = \begin{pmatrix}\mathbf{0} & \mathbf{0}\\ \mathbf{0} & \mathbf{I}\end{pmatrix},$$

see Mathai et al. [25,p.19]. We need the following traces:

$$\begin{aligned}
&\mathrm{tr}(\mathbf{AV}) = 0; \quad \mathrm{tr}(\mathbf{BV}) = \mathrm{tr}(\mathbf{V}) = 2\,\mathrm{tr}(\boldsymbol{\Sigma}); \quad \mathrm{tr}(\mathbf{CV}) = \mathrm{tr}(\boldsymbol{\Sigma}) = \mathrm{tr}(\mathbf{DV});\\
&\mathrm{tr}(\mathbf{AV})^2 = 2\,\mathrm{tr}(\boldsymbol{\Sigma}^2) = \mathrm{tr}(\mathbf{BV})^2; \quad \mathrm{tr}(\mathbf{CV})^2 = \mathrm{tr}(\boldsymbol{\Sigma}^2) = \mathrm{tr}(\mathbf{DV})^2;\\
&\mathrm{tr}(\mathbf{AVBV}) = 0 = \mathrm{tr}(\mathbf{AVCV}) = \mathrm{tr}(\mathbf{AVDV});\\
&\mathrm{tr}(\mathbf{AVBV})^2 = 2\,\mathrm{tr}(\boldsymbol{\Sigma}^4); \quad \mathrm{tr}(\mathbf{AVCV})^2 = 0 = \mathrm{tr}(\mathbf{AVDV})^2;\\
&\mathrm{tr}[(\mathbf{AV})^2(\mathbf{BV})] = 2\,\mathrm{tr}(\boldsymbol{\Sigma}^3); \quad \mathrm{tr}[(\mathbf{AV})(\mathbf{BV})^2] = 0; \quad \mathrm{tr}[(\mathbf{AV})^2(\mathbf{BV})^2] = 2\,\mathrm{tr}(\boldsymbol{\Sigma}^4);\\
&\mathrm{tr}[(\mathbf{AV})^2(\mathbf{CV})] = \mathrm{tr}(\boldsymbol{\Sigma}^3); \quad \mathrm{tr}[(\mathbf{AV})(\mathbf{CV})^2] = 0; \quad \mathrm{tr}[(\mathbf{AV})^2(\mathbf{CV})^2] = \mathrm{tr}(\boldsymbol{\Sigma}^4);\\
&\mathrm{tr}[(\mathbf{AV})^2(\mathbf{DV})] = \mathrm{tr}(\boldsymbol{\Sigma}^3); \quad \mathrm{tr}[(\mathbf{AV})(\mathbf{DV})^2] = 0; \quad \mathrm{tr}[(\mathbf{AV})^2(\mathbf{DV})^2] = \mathrm{tr}(\boldsymbol{\Sigma}^4).
\end{aligned}$$

Since many of the traces vanish, the moments of Lemma A.2 reduce to the following simple forms:

$$\begin{aligned}
\mathrm{E}(z) &= 0,\\
\mathrm{E}(z^2) &= 32\,\mathrm{tr}[(\mathbf{A}\boldsymbol{\Sigma})^2(\mathbf{B}\boldsymbol{\Sigma})^2] + 16[\mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma}\mathbf{B}\boldsymbol{\Sigma})^2 + \mathrm{tr}(\mathbf{B}\boldsymbol{\Sigma})\,\mathrm{tr}\{(\mathbf{A}\boldsymbol{\Sigma})^2(\mathbf{B}\boldsymbol{\Sigma})\}]\\
&\quad + 4\,\mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma})^2\,\mathrm{tr}(\mathbf{B}\boldsymbol{\Sigma})^2 + 2[\mathrm{tr}(\mathbf{B}\boldsymbol{\Sigma})]^2\,\mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma})^2.
\end{aligned} \tag{A3}$$

Using these results, we obtain the following moments:

$$\begin{aligned}
\mathrm{E}(\mathbf{Z}'\mathbf{AZ}\cdot\mathbf{Z}'\mathbf{BZ})^2 &= 96\,\mathrm{tr}(\boldsymbol{\Sigma}^4) + 64\,\mathrm{tr}(\boldsymbol{\Sigma}^3)\,\mathrm{tr}(\boldsymbol{\Sigma}) + 16[\mathrm{tr}(\boldsymbol{\Sigma}^2)]^2 + 16[\mathrm{tr}(\boldsymbol{\Sigma})]^2\,\mathrm{tr}(\boldsymbol{\Sigma}^2),\\
\mathrm{E}(\mathbf{Z}'\mathbf{AZ}\cdot\mathbf{Z}'\mathbf{CZ})^2 &= 32\,\mathrm{tr}(\boldsymbol{\Sigma}^4) + 16\,\mathrm{tr}(\boldsymbol{\Sigma}^3)\,\mathrm{tr}(\boldsymbol{\Sigma}) + 8[\mathrm{tr}(\boldsymbol{\Sigma}^2)]^2 + 4[\mathrm{tr}(\boldsymbol{\Sigma})]^2\,\mathrm{tr}(\boldsymbol{\Sigma}^2),\\
\mathrm{E}(\mathbf{Z}'\mathbf{AZ}\cdot\mathbf{Z}'\mathbf{DZ})^2 &= \mathrm{E}(\mathbf{Z}'\mathbf{AZ}\cdot\mathbf{Z}'\mathbf{CZ})^2.
\end{aligned}$$

which gives, from Equation (A2),

$$\mathrm{E}(A_kA_lA_{kl}^2) = 4\,\mathrm{tr}(\boldsymbol{\Sigma}^4) + 4\,\mathrm{tr}(\boldsymbol{\Sigma}^3)\,\mathrm{tr}(\boldsymbol{\Sigma}) + [\mathrm{tr}(\boldsymbol{\Sigma})]^2\,\mathrm{tr}(\boldsymbol{\Sigma}^2), \tag{A4}$$

so that $\mathrm{Cov}(A_kA_l, A_{kl}^2) = 4\,\mathrm{tr}(\boldsymbol{\Sigma}^4) + 4\,\mathrm{tr}(\boldsymbol{\Sigma}^3)\,\mathrm{tr}(\boldsymbol{\Sigma})$. Finally, Equation (A5) gives

$$\begin{aligned}
\mathrm{Cov}(C_2, C_3) &= \frac{1}{n^2(n-1)^2}\left[2n(n-1)\left\{4\,\mathrm{tr}(\boldsymbol{\Sigma}^4) + 4\,\mathrm{tr}(\boldsymbol{\Sigma}^3)\,\mathrm{tr}(\boldsymbol{\Sigma})\right\} + 4n(n-1)(n-2)\left\{2\,\mathrm{tr}(\boldsymbol{\Sigma}^3)\,\mathrm{tr}(\boldsymbol{\Sigma})\right\}\right]\\
&= \frac{8}{n(n-1)}[\mathrm{tr}(\boldsymbol{\Sigma}^4) + (n-1)\,\mathrm{tr}(\boldsymbol{\Sigma}^3)\,\mathrm{tr}(\boldsymbol{\Sigma})].
\end{aligned} \tag{A5}$$

Furthermore,

$$\begin{aligned}
\frac{\mathrm{Cov}(C_2, C_3)}{[\mathrm{tr}(\boldsymbol{\Sigma})]^2\,\mathrm{tr}(\boldsymbol{\Sigma}^2)} &= \frac{8}{n(n-1)}\left(\frac{\mathrm{tr}(\boldsymbol{\Sigma}^4)}{[\mathrm{tr}(\boldsymbol{\Sigma})]^2\,\mathrm{tr}(\boldsymbol{\Sigma}^2)} + \frac{(n-1)\,\mathrm{tr}(\boldsymbol{\Sigma}^3)}{\mathrm{tr}(\boldsymbol{\Sigma})\,\mathrm{tr}(\boldsymbol{\Sigma}^2)}\right)\\
&\le \frac{8}{n-1} = O\left(\frac{1}{n}\right),
\end{aligned}$$

by the Cauchy–Schwarz inequality. Now,

$$\mathrm{Cov}(C_3, C_1) = \frac{1}{n^2(n-1)}\underbrace{\sum_{k=1}^{n}\sum_{l=1}^{n}\sum_{m=1}^{n}}_{k\neq(l,m),\,l\neq m}\mathrm{Cov}(A_k, A_{lm}^2), \tag{A6}$$

where $\mathrm{Cov}(A_k, A_{lm}^2) = \mathrm{E}(A_kA_{lm}^2) - \mathrm{tr}(\boldsymbol{\Sigma})\,\mathrm{tr}(\boldsymbol{\Sigma}^2)$, from Theorem A.1. The covariance is zero for $k \neq l \neq m$. The other two cases, $k = l \neq m$ and $k = m \neq l$ yield the same result. Then, for $k = l \neq m$, we have

$$\begin{aligned}
\mathrm{E}(A_kA_{lm}^2) &= \mathrm{E}(\mathbf{X}_k'\mathbf{X}_k\mathbf{X}_k'\mathbf{X}_m\mathbf{X}_k'\mathbf{X}_m) = \mathrm{E}[\mathrm{tr}(\mathbf{X}_k'\mathbf{X}_k\mathbf{X}_k'\mathbf{X}_m\mathbf{X}_m'\mathbf{X}_k)]\\
&= \mathrm{E}[\mathrm{tr}(\mathbf{X}_k\mathbf{X}_k'\mathbf{X}_k\mathbf{X}_k'\mathbf{X}_m\mathbf{X}_m')] = \mathrm{tr}[\mathrm{E}(\mathbf{X}_k\mathbf{X}_k'\mathbf{X}_k\mathbf{X}_k')\boldsymbol{\Sigma}]\\
&= \mathrm{E}(\mathbf{X}_k'\mathbf{X}_k\mathbf{X}_k'\boldsymbol{\Sigma}\mathbf{X}_k) = 2\,\mathrm{tr}(\boldsymbol{\Sigma}^3) + \mathrm{tr}(\boldsymbol{\Sigma})\,\mathrm{tr}(\boldsymbol{\Sigma}^2),
\end{aligned}$$

from Lemma A.2, so that $\mathrm{Cov}(A_k, A_{lm}^2) = 2\,\mathrm{tr}(\boldsymbol{\Sigma}^3)$, and from Equation (A8)

$$\mathrm{Cov}(C_3, C_1) = \frac{1}{n^2(n-1)}\left[2n(n-1)\{2\,\mathrm{tr}(\boldsymbol{\Sigma}^3)\}\right] = \frac{4}{n}\,\mathrm{tr}(\boldsymbol{\Sigma}^3), \tag{A7}$$

where, by the Cauchy–Schwarz inequality

$$\frac{\mathrm{Cov}(C_3, C_1)}{\mathrm{tr}(\boldsymbol{\Sigma}^2)\,\mathrm{tr}(\boldsymbol{\Sigma})} = \frac{4}{n}\left(\frac{\mathrm{tr}(\boldsymbol{\Sigma}^3)}{\mathrm{tr}(\boldsymbol{\Sigma}^2)\,\mathrm{tr}(\boldsymbol{\Sigma})}\right) \le \frac{4}{n} = O\left(\frac{1}{n}\right).$$

Finally,

$$\mathrm{Cov}(C_1, C_2) = \frac{1}{n^2(n-1)}\underbrace{\sum_{k=1}^{n}\sum_{l=1}^{n}\sum_{m=1}^{n}}_{k\neq(l,m),l\neq m}\mathrm{Cov}(A_k, A_lA_m), \tag{A8}$$

where $\mathrm{Cov}(A_k, A_lA_m) = \mathrm{E}(A_kA_lA_m) - [\mathrm{tr}(\boldsymbol{\Sigma})]^3$, from Theorem A.1. Then, working on the same lines as for $\mathrm{Cov}(C_3, C_1)$, it can be shown that

$$\mathrm{Cov}(C_1, C_2) = \frac{4}{n}\,\mathrm{tr}(\boldsymbol{\Sigma}^2)\,\mathrm{tr}(\boldsymbol{\Sigma}), \tag{A9}$$

so that, again by the Cauchy–Schwarz inequality,

$$\frac{\mathrm{Cov}(C_1, C_2)}{\mathrm{tr}(\boldsymbol{\Sigma})[\mathrm{tr}(\boldsymbol{\Sigma})]^2} \le \frac{4}{n} = O\left(\frac{1}{n}\right).$$

Appendix 3. Proofs of Theorems 8 and 10

Let $R_1 = C_1/\mathrm{tr}(\boldsymbol{\Sigma})$ and $R_2 = C_2/[\mathrm{tr}(\boldsymbol{\Sigma})]^2$. Using the results of Lemma 2, it can be immediately shown that $\mathrm{E}(R_1) = 1$, $\mathrm{E}(R_2) = 1$, $\mathrm{Var}(R_1) \le 2/n$, and $\mathrm{Var}(R_2) \le 8/(n-1)$, so that, by Chebychev's inequality, $R_1 \xrightarrow{\mathcal{P}} 1$ and $R_2 \xrightarrow{\mathcal{P}} 1$, as $n, p \to \infty$, as required. Then, to prove Theorems 8 and 10, we only need to show the asymptotic normality of Equation (10). To use Lehmann's theorem, let

$$h(\mathbf{X}_1, \ldots, \mathbf{X}_m) = \frac{1}{\mathrm{tr}(\boldsymbol{\Sigma}^2)}A_{kl}^2 = \frac{1}{\mathrm{tr}(\boldsymbol{\Sigma}^2)}\mathbf{X}_k'\mathbf{X}_l\mathbf{X}_k'\mathbf{X}_l = \frac{1}{\mathrm{tr}(\boldsymbol{\Sigma}^2)}\mathbf{X}_k'\mathbf{X}_l\mathbf{X}_l'\mathbf{X}_k,$$

with $h_c(\mathbf{X}_1, \ldots, \mathbf{X}_c) = \mathrm{E}(h(\mathbf{X}_1, \ldots, \mathbf{X}_m)|\mathbf{X}_1 = \mathbf{x}_1, \ldots, \mathbf{X}_c = \mathbf{x}_c)$, $1 \le c \le m$. Now, with $m = 2$ and $c = 1, 2$, we have

$$h_1(\mathbf{X}_k) = \frac{1}{\mathrm{tr}(\boldsymbol{\Sigma}^2)}\mathbf{x}_k'\boldsymbol{\Sigma}\mathbf{x}_k,$$

and $h_2(\mathbf{X}_k, \mathbf{X}_l) = h(\cdot)$. Then, $\mathrm{E}(h_c(\cdot)) = 1 = \mathrm{E}(U_n)$, $c = 1, 2$, and

$$\xi_1 = \frac{2\,\mathrm{tr}(\boldsymbol{\Sigma}^4)}{\left[\mathrm{tr}(\boldsymbol{\Sigma}^2)\right]^2}, \quad \xi_2 = \frac{6\,\mathrm{tr}(\boldsymbol{\Sigma}^4) + 2[\mathrm{tr}(\boldsymbol{\Sigma}^2)]^2}{[\mathrm{tr}(\boldsymbol{\Sigma}^2)]^2},$$

using the results of Theorem A.1 and Lemma 2, so that, (see Lehmann [15], Equation 6.1.17, p. 368)

$$\begin{aligned}\mathrm{Var}(U_n) &= \binom{n}{2}^{-1}\sum_{c=1}^{2}\binom{2}{c}\binom{n-2}{2-c}\xi_c \\ &= \frac{2}{n(n-1)}[2(n-2)\xi_1 + \xi_2] \\ &= \frac{4}{n(n-1)[\mathrm{tr}(\boldsymbol{\Sigma}^2)]^2}((2n-1)\,\mathrm{tr}(\boldsymbol{\Sigma}^4) + [\mathrm{tr}(\boldsymbol{\Sigma}^2)]^2).\end{aligned}$$

Furthermore, by the Cauchy–Schwarz inequality and Assumption 7, $0 < \xi_1 \le 2$ and $0 < \xi_2 \le 8$, so that, by Lehmann's theorem, $(U_n - 1)/\mathrm{Var}(U_n) \xrightarrow{\mathcal{D}} N(0, 1)$, as $n, p \to \infty$, under Assumptions 4–7. This leads to the proof of Theorem 8, by Slutsky's theorem [20, Theorem 2.13, p. 30].

We note that, under the null hypothesis, $\mathrm{Var}(nU_n) \to 4(2/c + 1)$, by Assumption 6, as $p \to \infty$, which gives the null distribution of T_1. This completes the proof of Theorem 8 and Corollary 9, and essentially also of Theorem 10 and Corollary 11, since both test statistics share the same kernel.